Customer Support Ticket Analysis

March 31, 2026

1 Dataset Overview

The dataset consists of 8469 customer support tickets and 17 features, including both structured and unstructured data. All columns are fully populated except for a subset of resolution-related variables. The dataset contains:

- 2 numerical features e.g. Customer Age
- 1 numerical target variable: Customer Satisfaction Rating.
- 14 categorical or text-based features

2 Data Quality Analysis

2.1 Missing Values

A significant proportion of missing values is observed in the dataset, particularly in the following variables:

- Customer Satisfaction Rating (67.3%)
- Time to Resolution (67.3%)
- Resolution (67.3%)
- First Response Time (33.3%)

These missing values are not random. They correspond to tickets that have not yet been resolved. This is supported by the fact that the number of non-missing values closely matches the number of closed tickets.

3 Exploratory Data Analysis

3.1 Numerical Features

Customer age ranges from 18 to 70, with an average of approximately 44 years and a standard deviation of 15.3. The distribution appears relatively balanced, with no extreme outliers.

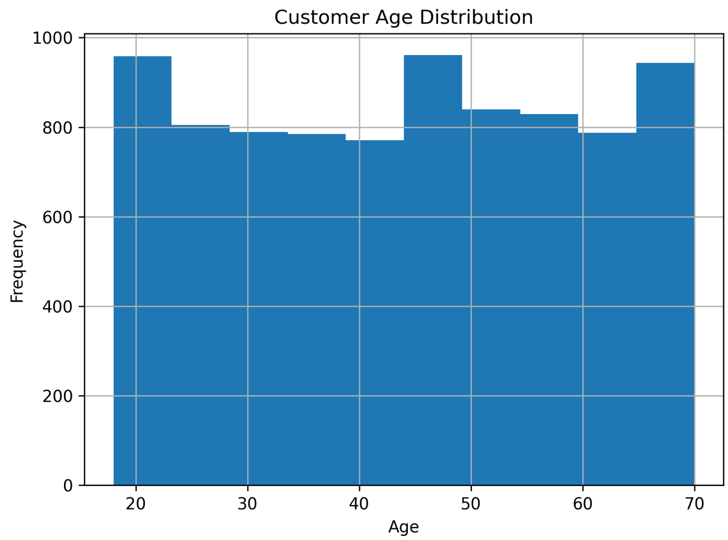

Figure 1: Distribution of customer age

Customer satisfaction ratings range from 1 to 5, with an average value of 2.99. This suggests that overall satisfaction is moderate.

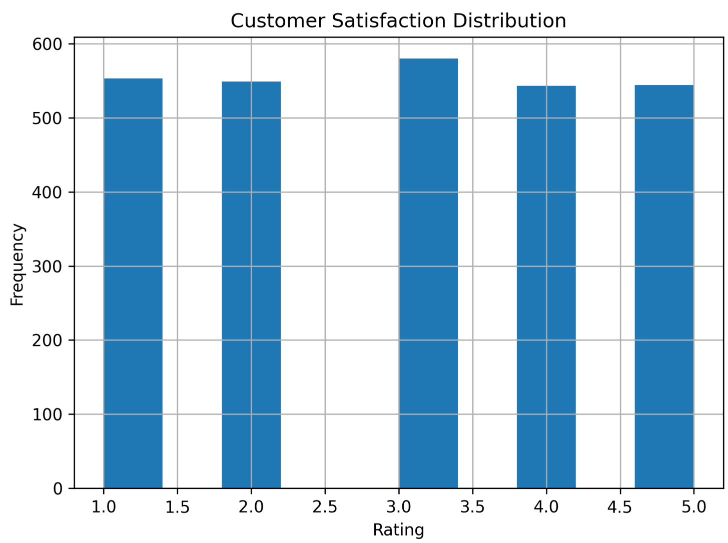

Figure 2: Distribution of customer satisfaction ratings

3.2 Categorical Features

Ticket types are evenly distributed across five categories, including refund requests, technical issues, cancellation requests, product inquiries, and billing inquiries. Each category contains approximately 1600–1750 tickets.

Similarly, ticket priority levels: low, medium, high, and critical, are uniformly distributed, with each category containing around 2000 instances.

Ticket channels (email, phone, chat, and social media) are also evenly represented, suggesting that customers use multiple communication channels consistently.

Ticket status is distributed across three categories:

- Pending Customer Response (2881)
- Open (2819)
- Closed (2769)

4 Relationship Analysis

4.1 Priority and Satisfaction

An analysis of ticket priority and customer satisfaction reveals a clear trend: higher priority tickets tend to have lower satisfaction scores.

Specifically:

- Low priority tickets have the highest average satisfaction (3.05)
- Critical tickets have the lowest average satisfaction (2.96)

This suggests that more severe issues are associated with poorer customer experiences.

4.2 Channel and Ticket Type

Cross-tabulation of ticket channel and ticket type shows relatively consistent distributions across channels. However, certain patterns emerge, such as a slightly higher proportion of technical issues in social media and chat channels.

This indicates that different communication channels may be associated with different types of customer issues.

5 NLP-Oriented Exploratory Data Analysis

5.1 Text Preprocessing

The ticket descriptions were preprocessed to improve text quality. This included:

- Converting text to lowercase
- Removing punctuation and non-alphabetic characters
- Removing stopwords
- Removing template placeholders (e.g., {product_purchased})

This step is essential to ensure that subsequent analysis reflects meaningful content rather than noise.

5.2 Word Frequency Analysis

After preprocessing, word frequency analysis was conducted to identify commonly occurring terms in customer complaints.

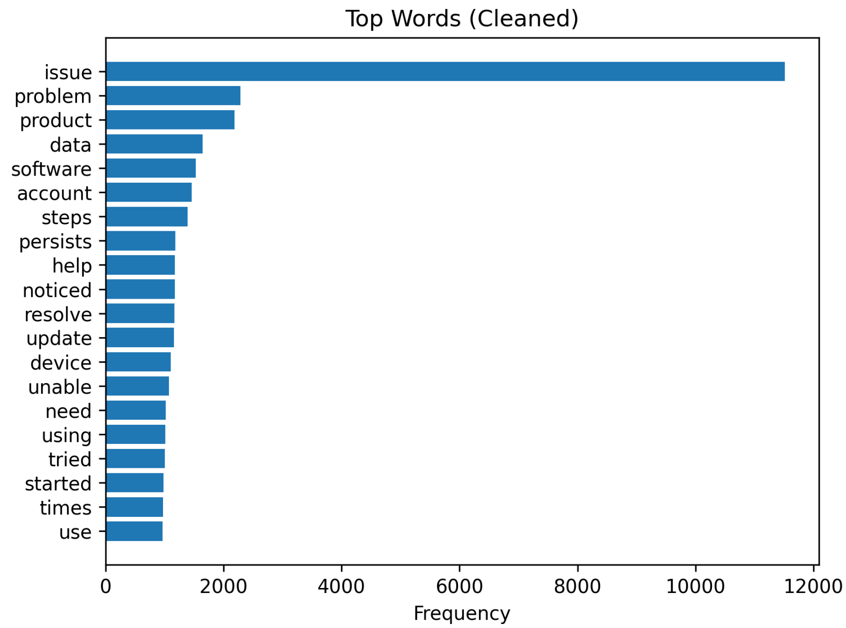

Figure 3: Top 20 most frequent words after cleaning

Compared to the raw text, the cleaned results provide more meaningful insights into customer concerns.

5.3 Bigram Analysis

To capture more context, bigram analysis (pairs of consecutive words) was performed.

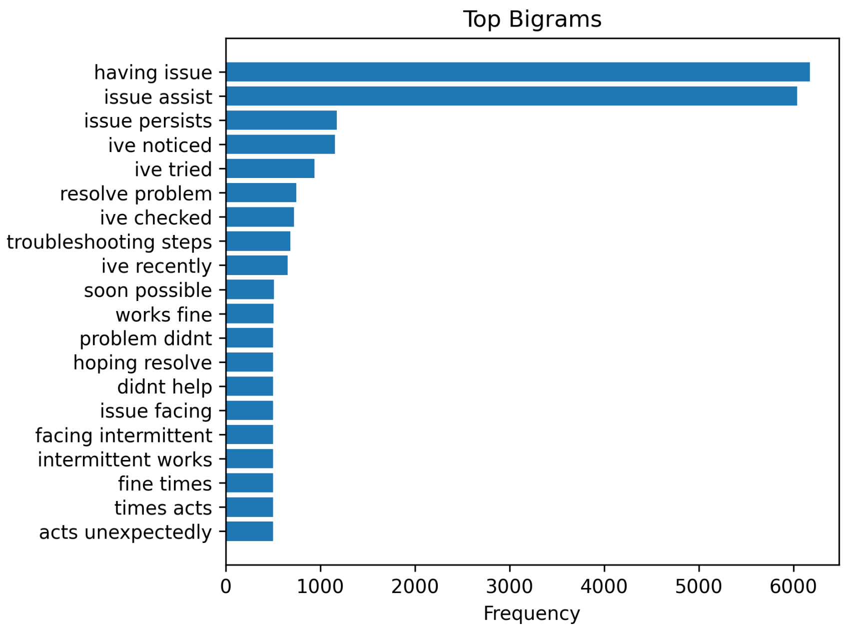

Figure 4: Top 20 bigrams

Common phrases such as "refund request" and "technical issue" provide clearer representations of customer problems than individual words.

5.4 Text Length Analysis

Text length was measured in terms of word count rather than character count to better reflect information content.

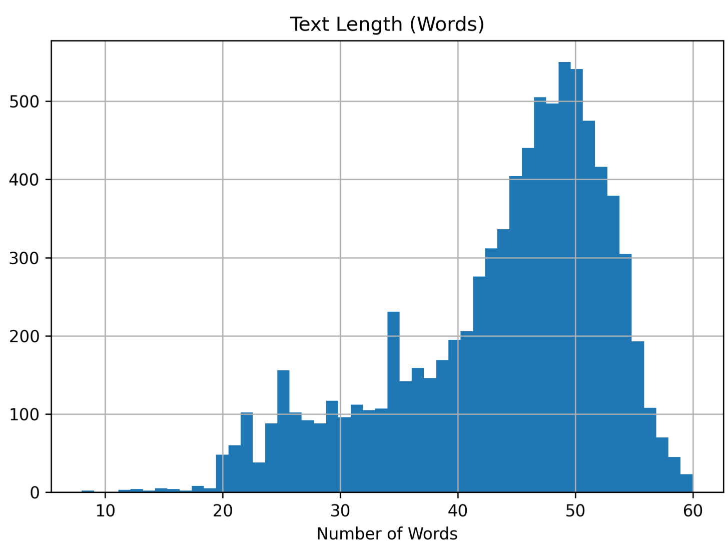

Figure 5: Distribution of text length (in words)

The distribution indicates that most tickets contain a moderate amount of detail, suitable for

NLP analysis.

5.5 Analysis by Ticket Type

Word frequency was further analysed across different ticket types.

Key observations include:

- Refund requests frequently include words such as “refund” and “return”
- Technical issues commonly include terms like “error” and “not working”
- Billing inquiries often mention “charge” and “payment”

This demonstrates that different ticket categories exhibit distinct linguistic patterns, which can be leveraged for classification and clustering.

6 Conclusion